Search and Consideration Sets

October, 2026

Where We Are

- In *Demand and Welfare* and *Dynamic Demand* every consumer knows every product's price and characteristics, and chooses from the full choice set.
- This lecture drops that assumption. Consumers must *become aware of* and *search* products before they can buy them.
- Two reasons an IO economist should care:
  - **Measurement.** If consumers only compare a handful of products, a full-information model mis-reads limited substitution as low price sensitivity, and gets elasticities, markups and merger effects wrong.
  - **Market power.** Search frictions are themselves a source of market power (Diamond 1971), and they rationalize price dispersion for identical goods.

Introduction

- The classical search literature is motivated by substantial price dispersion for similar, even identical, goods (Stigler 1961).
- The same logic applies to any product attribute the consumer must discover, which challenges the "full information" assumption behind standard demand estimation.
- Three elements of the process
  - the consumer's search process;
  - the formation of a consideration set;
  - a model of choice, given the consideration set.
- Stigler meets the marketing evidence
  - consumers consider a surprisingly small number of alternatives (3–5);
  - advertising/promotion works foremost through awareness (with little consumption utility).

Road-map

1. **Consideration sets without search.** Awareness is a statistical process shifted by advertising. (Goeree 2008)

2. **Simultaneous (fixed-sample) search.**
   - Theory: Stigler, the Rothschild critique, the Diamond paradox, Burdett–Judd.
   - Search costs from price data alone. (Hong & Shum 2006)
   - Awareness $\rightarrow$ consideration $\rightarrow$ choice with micro data. (Honka, Hortaçsu & Vitorino 2017)

3. **Sequential search.**
   - Weitzman (1979) and reservation utilities.
   - Search from view-rank data. (Kim, Albuquerque & Bronnenberg 2010)
   - Search and product differentiation in equilibrium. (Hortaçsu & Syverson 2004)

4. **Sequential search inside a BLP-style demand system.** (Moraga-González, Sándor & Wildenbeest 2023)

Framework (Honka, Hortaçsu & Wildenbeest 2019)

- Consumer i's indirect utility for variety j

$$u_{ij} = \alpha_j + X_{ij}\beta + \gamma p_{ij} + \epsilon_{ij}, \qquad \epsilon_{ij} \sim F_j.$$

  Special case $u_{ij} = -p_{ij}$, $F_j = F$: homogeneous goods, search is over prices only.

- Maintained assumptions
  - discrete choice;
  - rational expectations: consumers know F_j (but not the draw);
  - uncertainty about u_{ij} is resolved upon search;
  - the first search is free (for convenience).

- What differs across papers: *what* is unknown before search (p_{ij}, ϵ_{ij}, or both), the search *protocol* (simultaneous vs. sequential), and the *data* (prices only, shares, consideration sets, search paths).

Consideration Sets

- In principle, one does not need to model search to estimate demand if the consideration set *were observed*.

- Earlier literature
  - Only choice data are observed, so the consideration set is recovered through functional form.
  - *No distinction between "awareness" and "consideration"*: the awareness set is formed statistically, but can be shifted by advertising/promotion.
  - Implication: *no effort* is needed to discover ϵ_{ij} or p_{ij}.
  - Goeree (2008)'s "choice set" is the leading example.

- Where search adds something
  - discovering ϵ_{ij} or p_{ij} is costly $\Rightarrow$ an economic model disciplines how the consideration set is formed;
  - the search cost is itself an object of interest (a source of market power).

Goeree (2008): Model

- A simplified version without random coefficients

$$u_{ij} = \delta_j + \mu_{ij} + \epsilon_{ij}, \quad \mu_{ij} = \alpha \ln(y_i - p_j) + x_j' \Omega D_i, \quad \delta_j = x_j' \beta + \xi_j$$

- Consumer i faces a choice set S that contains an incomplete set of varieties

$$\Pr(S) = \prod_{l \in S} \phi_{il} \prod_{k \notin S} (1 - \phi_{ik}), \qquad \phi_{ij} = \frac{\exp(\gamma_j + \lambda_{ij})}{1 + \exp(\gamma_j + \lambda_{ij})}$$

- The individual purchase probability is then

$$s_{ij} = \sum_{S \ni j} \Pr(S) \frac{\exp(\delta_j + \mu_{ij})}{y_i^{\alpha} + \sum_{k \in S} \exp(\delta_k + \mu_{ik})}$$

- Advertising technology

$$\gamma_j = a_j'(\psi + \rho a_j + i_m \psi_f) + \nu x_j^{age}$$
$$\lambda_{ij} = a_j'(\Upsilon_m D_i^s \zeta) + \tilde{D}_i \tilde{\lambda}$$

ψ_f is a firm effect, x_j^{age} is product age, $\Upsilon_m D_i^s$ is individual media exposure, and $\tilde{\lambda}$ is individual awareness absent advertising.

Goeree (2008): Aggregation and Simulation

- Aggregate to product j's market share

$$s_j = \int s_{ij}\, dG_{y,D}(y, D)$$

- The outer integral is approximated by simulating $G(\cdot)$ as in BLP, i.e. random draws of consumers i.

- The sum over S suffers a "curse of dimensionality" (2^J sets)
  - Again simulate: draw a J-dimensional Bernoulli vector $b_i(\theta)$ for each i, given $\phi_{ij}(\theta)$ and $U[0, 1]$ draws.
  - $\hat{s}_{ij} = \frac{1}{R} \sum_r s_{ijr}$.

- The choice set S is formed statistically and shifted by a_j (advertising). How does the paper identify this process?

Goeree (2008): Identification

- Identification relies on functional form plus exclusion.
- Advertising and PC age do not enter utility directly: variation in s_j across products and over time identifies ψ, ρ, ν.

- Supplement with micro data
  - Simmons survey: 20,000 households, media habits (exposure to advertising), product usage, and demographics.
  - Simmons also asks whether the household purchased a new PC and from which manufacturer.

- Two sets of micro moments
  - $E[B_i \mid D_i^s, y_i]$: the expected choice of manufacturer for each household type in the data (à la Petrin 2002);
  - $E[H_{im} \mid D_i^s, \Upsilon_m]$: individual media exposure, estimated *directly* from Simmons. Given Υ_m, the key parameter is then ζ.

Simultaneous Search

- Simultaneous (fixed-sample) search: the consumer commits up front to searching a set S

$$S_i = \arg\max_S \left[E\left[\max_{j \in S} u_{ij} \right] - \sum_{j \in S} c_{ij} \right]$$

- A severe "curse of dimensionality": $2^J - 1$ candidate search sets.

- Needs additional structure
  - Homogeneous goods: Stigler (1961), relying primarily on the observed price distribution.
  - Chade & Smith (2006): when options can be ranked by *expected utility*, the consumer searches the top k and only the size k must be optimized (used by Honka 2014 and Honka, Hortaçsu & Vitorino 2017).

Simultaneous Search: Earlier Theories

- A **unit** continuum of price-setting firms selling an identical product at marginal cost mc
- Mass μ of consumers
- Consumer indirect utility $V(p, M) = v(p) + M$; by Roy's identity $q(p) \equiv -v'(p)$
- Search cost c per price quote
- Consumers know the distribution of posted prices, but not the price charged by any particular seller

Stigler (1961)

- Consumers purchase $K \geq 1$ units of the product ($-v'(p) = K$).
- Fixed-sample search of size n.
- The distribution of prices is *exogenous*: $F(p)$ on $[\underline{p}, \bar{p}]$.
- The consumer minimizes expected total cost

$$E[C] = K \int_{\underline{p}}^{\bar{p}} p \, dF^n_{\min}(p) + cn = K \int_{\underline{p}}^{\bar{p}} np(1 - F(p))^{n-1} f(p) \, dp + cn$$

- The optimal n^* implies the transaction price distribution $F^{n^*}_{\min}(p) \equiv 1 - (1 - F(p))^{n^*}$.
- A representative firm's (from a unit mass) expected demand is

$$Q(p) = \mu \times \underbrace{n^*(1 - F(p))^{n^*-1}}_{\text{searched and lowest price}} \times K$$

- Expected transaction prices and total costs are lower when prices are *more dispersed*.

The Rothschild Critique

- Fixed-sample search need not be optimal $\rightarrow$ sequential search.
- **More importantly**: the posted price distribution is not derived from *firm optimization*.
- Indeed, absent cost heterogeneity, firms have no incentive to post different prices.
- We see this argument at its extreme in the "Diamond paradox."

The Diamond Paradox: Setup

- Identical downward-sloping demand
- Consumers can search optimally and sequentially
- Search cost $c > 0$ per search
- At the monopoly price p_m, consumers still have positive surplus: $v(p_m) > c$
- What is the equilibrium of this market?

The Diamond Paradox: Result

Diamond (1971): the unique equilibrium has every firm charging the monopoly price p_m.

- Consumers obtain exactly one quote and buy.

Why no firm undercuts:

- Suppose the lowest price in the market is some $p < p_m$. The firm charging it can raise its price by (almost) c and still keep its customers: searching again saves them less than c.
- Repeat the argument from the new lowest price: prices ratchet up until p_m.
- No firm wants to go above p_m (it lowers profit from its captive customers).

The presence of search frictions is a source of market power.

- The Diamond paradox is extreme: consumers do not search at all.
- And there is no price dispersion, contrary to the facts that motivated Stigler.

Burdett & Judd (1983): Search Behaviour

- Adds *ex post* heterogeneity in consumer information and *mixed* pricing strategies.
- Identical firms and identical unit-demand consumers with valuation v; fixed-sample search as in Stigler.
- Start from a price distribution $F(p)$; let θ_n be the fraction of consumers who optimally sample n firms.
- The expected benefit of sampling one more firm is

  $$E[B^{(n)}] = E[p_{\min}^{(n-1)}] - E[p_{\min}^{(n)}],$$

  and n is optimal iff $E[B^{(n+1)}] < c \leq E[B^{(n)}]$.

Burdett & Judd (1983): Equilibrium

- If every consumer obtains two or more quotes ($\theta_1 = 0$), firms price at marginal cost (Bertrand), and then nobody would pay c for a second quote. Contradiction.
- So $\theta_1 > 0$; but $\theta_1 = 1$ is the Diamond outcome, with no reason to post anything but v.
- Since $E[B^{(n)}]$ is decreasing in n and consumers are identical, in a dispersed equilibrium they are indifferent between one and two quotes:

$$E[B^{(2)}] = c, \qquad \theta_1 \equiv \theta, \quad \theta_2 = 1 - \theta.$$

- A firm is sampled by $\mu\theta$ one-quote consumers and by $2\mu(1-\theta)$ two-quote consumers, and wins the latter only if it is cheaper:

$$E[\pi \mid p = v] = (v - mc)\,\mu\theta$$
$$E[\pi \mid p < v] = (p - mc)\,\mu\big[\theta + 2(1-\theta)(1 - F(p))\big]$$

- Indifference across the support yields the equilibrium price distribution

$$F(p) = 1 - \frac{\theta}{2(1-\theta)}\,\frac{v-p}{p-mc}, \qquad p \in \left[\tfrac{\theta v + 2(1-\theta)\,mc}{2-\theta},\ v\right]$$

Hong & Shum (2006)

- An empirical version of Burdett–Judd, with a non-degenerate search cost distribution that micro-founds the mixed strategy.
- Question: how can the *price distribution alone* be used to estimate search costs?
- Continuum of firms and consumers
  - F_p: observed price distribution
  - F_c: search cost distribution (per store)
  - mc: common retailer cost
- Two consumer search protocols
  - simultaneous search (below)
  - sequential search (in the paper; not covered here)
- The first quote is free.

Hong & Shum (2006): Consumers

- A consumer with search cost c who samples n stores incurs

$$c(n-1) + E[\min(p_1, \ldots, p_n)] = c(n-1) + \int_{\underline{p}}^{\bar{p}} p\, n(1 - F_p(p))^{n-1} f_p(p)\, dp$$

  using the order statistic $F^{(n)}_{p_{\min}}(p) = 1 - [1 - F_p(p)]^n$.

- Optimal n is decreasing in c, so search behaviour is characterized by cut-offs (à la Burdett–Judd)

$$c_n = E[\min(p_1, \ldots, p_n)] - E[\min(p_1, \ldots, p_{n+1})], \qquad c_0 \equiv \infty.$$

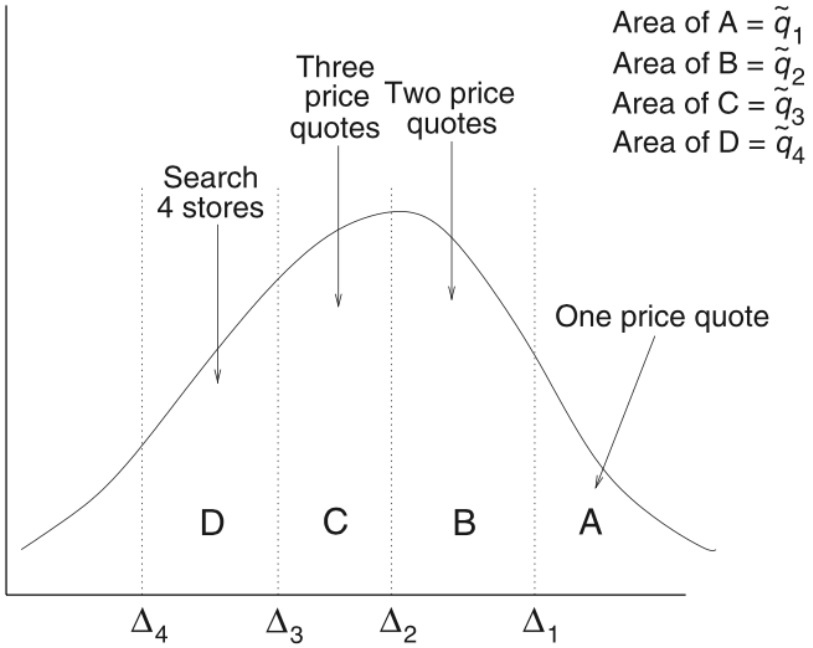
- Consumers with $c \in (c_n, c_{n-1}]$ sample n stores; the fraction doing so is

$$\tilde{q}_n = F_c(c_{n-1}) - F_c(c_n).$$

Hong & Shum (2006): Search Cost Cut-offs

FIGURE 2

IDENTIFICATION SCHEME FOR SEARCH-COST DISTRIBUTION IN NONSEQUENTIAL-SEARCH MODEL



Hong & Shum (2006): Firms and Estimation

- A firm charging p earns

$$\Pi(p) = (p - mc) \sum_{k=1}^{\infty} \tilde{q}_k \, k \, (1 - F_p(p))^{k-1}$$

- Mixed strategy: firms are indifferent across all prices in the support, and at $\bar{p}$ only one-quote consumers buy

$$(\bar{p} - mc) \, \tilde{q}_1 = (p - mc) \sum_{k=1}^{\infty} \tilde{q}_k \, k \, (1 - F_p(p))^{k-1}, \quad \forall p \in [\underline{p}, \bar{p})$$

- With N observed prices and $\hat{F}_p$, this is a system of $N - 1$ equations in $mc, \tilde{q}_1, \ldots, \tilde{q}_K$; solvable for a maximal search size $K < N - 1$.

- Then use $\tilde{q}_n = F_c(c_{n-1}) - F_c(c_n)$, with the c_n computed from $\hat{F}_p$, to trace out (or fit a parametric) search cost distribution.

Hong & Shum (2006): Results

TABLE 2 **Search-Cost Distribution Estimates for Nonsequential-Search Model**

Product	K[a]	M[b]	$\tilde{q}_1$[c]	$\tilde{q}_2$	$\tilde{q}_3$	Selling Cost r	MEL Value
Parameter estimates and standard errors: nonsequential-search model							
Stokey-Lucas	3	5	.480 (.170)	.288 (.433)		49.52 (12.45)	102.62
Lazear	4	5	.364 (.926)	.351 (.660)	.135 (.692)	27.76 (8.50)	84.70
Billingsley	3	5	.633 (.944)	.309 (.310)		69.73 (68.12)	199.70
Duffie	3	5	.627 (1.248)	.314 (.195)		35.48 (96.30)	109.13

Search-cost distribution estimates

	Δ_1	$F_c(\Delta_1)$	Δ_2	$F_c(\Delta_2)$	Δ_3	$F_c(\Delta_3)$
Stokey-Lucas	2.32	.520	.68	.232		
Lazear	1.31	.636	.83	.285	.57	.150
Billingsley	2.90	.367	2.00	.058		
Duffie	2.41	.373	1.42	.059		

[a] Number of quantiles of search cost F_c that are estimated (see equation (5)). In practice, we set K and M to the largest possible values for which the parameter estimates converge. All combinations of larger K and/or larger M resulted in estimates that either did not converge or did not move from their starting values (suggesting that the parameters were badly identified).

[b] Number of moment conditions used in the empirical likelihood estimation procedure (see equation (17)).

[c] For each product, only estimates for $\tilde{q}_1, \ldots, \tilde{q}_{K-1}$ are reported; $\tilde{q}_K = 1 - \sum_{k=1}^{K-1} \tilde{q}_k$.

[d] Indifferent points Δ_k computed as $Ep_{(1:k)} - Ep_{(1:k+1)}$ (the expected price difference from having k versus $k+1$ price quotes), using the empirical price distribution. Including shipping and handling charges.

Honka, Hortaçsu & Vitorino (2017): Setup

- Brings *differentiated products* into fixed-sample search, with consumer-level data that separate awareness (as in Goeree), consideration (search), and choice.
- Focus: the role of advertising in US retail banking.
- Awareness a_{ij} for individual i and bank j

$$a_{ij} = \zeta_{0j} + \zeta_{1j} adv_j + \zeta_{2j} D_i + \zeta_{3j} f(b_{ij}) + \xi_{ij}$$

where $f(b_{ij})$ captures branch proximity.

- Utility of consumer i for bank j

$$u_{ij} = \alpha_j + \beta_1 p_{ij} + \beta_2 adv_j + \beta_3 f(b_{ij}) + \epsilon_{ij}$$

- adv_j, b_{ij} and ϵ_{ij} are known to the consumer; she must search to learn p_{ij}, at cost c per quote. Prices are EV distributed with location η and scale τ.
- So, given ϵ_{ij}, u_{ij} is EV with location $a^u_{ij} = \alpha_j + \beta_1 \eta + \beta_2 adv_j + \beta_3 f(b_{ij}) + \epsilon_{ij}$ and a common scale g (proportional to $\tau|\beta_1|$): same scale, different locations.

The Chade–Smith Ranking

- Chade & Smith (2006): when the utility distributions are ranked by first-order stochastic dominance (here: same scale, different location), consumer i can rank the options by expected indirect utility

  $$E[u_{ij}] = a^u_{ij} + g\,\gamma_{\text{Euler}}, \qquad \forall j \in A_i.$$

  Let R_{ik} be the set of the top k options for i.

- The consumer chooses the *size* k of the search set to maximize

  $$\max_k \;\; \Gamma_{ik} = E\left[\max_{j \in R_{ik}} u_{ij}\right] - kc,$$

  which is in closed form under EV.

- Given the consideration set $S_i \subseteq A_i$, she buys $j = \arg\max_{j \in S_i} u_{ij}$.

- The joint probability of (A_i, S_i, j) decomposes as

  $$P_{i,A_i} \times \int P_{ij|S_i,\epsilon_i}\, P_{i,S_i|A_i,\epsilon_i}\, dF_\epsilon.$$

- For a given draw of ϵ the probabilities are 0/1 (non-smooth); the paper uses a kernel-smoothed frequency simulator (McFadden 1989).

Honka, Hortaçsu & Vitorino (2017): Estimation

- Data: A_i observed, S_i observed (but not its ranking), choice j observed; the price distribution is recovered from observed quotes.
- **Awareness stage.** ξ_{ij} is logistic, so $j \in A_i$ with probability

$$\frac{\exp(\zeta_{0j} + \zeta_{1j} adv_j + \zeta_{2j} D_i + \zeta_{3j} f(b_{ij}))}{1 + \exp(\zeta_{0j} + \zeta_{1j} adv_j + \zeta_{2j} D_i + \zeta_{3j} f(b_{ij}))}$$

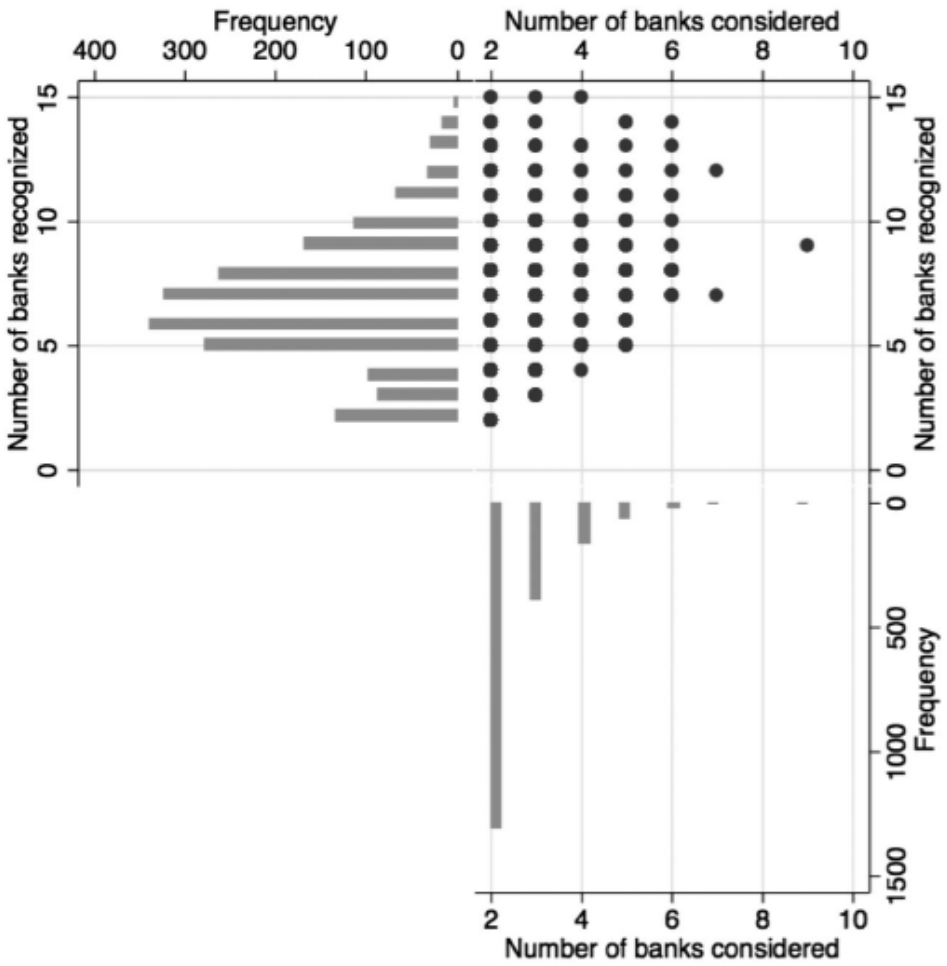
- **Consideration stage.** For simulated ϵ_i, compute $P_{i,S_i|A_i,\epsilon_i}$: every searched option ranks above every unsearched one, and $|S_i|$ is the optimal size

$$\Pr\left(\min_{j \in S_i} E[u_{ij}] \geq \max_{j' \in A_i \setminus S_i} E[u_{ij'}] \ \cap \ \Gamma_{i|S_i|} \geq \Gamma_{ik'} \ \forall k' \neq |S_i|\right)$$

- **Choice stage.** Compute $P_{ij|S_i,\epsilon_i}$. Although ϵ is EV, it has been *selected on* at the consideration stage, so there is no closed form.
- Both stages are averaged over the simulated ϵ_i.

Honka, Hortaçsu & Vitorino (2017): Data

AWARENESS VERSUS CONSIDERATION (SHOPPERS ONLY)



This figure shows the distribution of awareness and consideration set sizes for shoppers in our data and the relationship between the sizes of the two types of sets.

Full Information vs. Limited Information

- Utility estimates: full information (left) vs. limited information (baseline, right).
- Advertising works mainly through awareness, which enlarges consideration sets and so is pro-competitive.

Other Parameters		
Primary bank	3.020**	0.382**
	(0.061)	(0.030)
Interest rates	−0.575	2.104**
	(0.526)	(0.553)
Advertising	0.038**	0.015*
	(0.015)	(0.009)
Advertising residual	−0.043	−0.015
	(0.039)	(0.022)
Bank branches, N=1 (Y/N)	1.972**	0.504**
	(0.154)	(0.091)
Bank branches, $3 \geq N \geq 2$ (Y/N)	1.949**	0.655**
	(0.175)	(0.081)
Bank branches, $7 \geq N \geq 4$ (Y/N)	2.225**	0.796**
	(0.163)	(0.083)

(Continued)

Sequential Search: Weitzman (1979)

- Dynamic program in the best utility found so far $\tilde{u}_i$ and the set of unexplored options $\bar{S}_i$:

$$W(\tilde{u}_i, \bar{S}_i) = \max\left\{\tilde{u}_i,\ \max_{j \in \bar{S}_i}\left[-c_{ij} + F_j(\tilde{u}_i)\, W(\tilde{u}_i, \bar{S}_i \setminus \{j\}) + \int_{\tilde{u}_i}^{\infty} W(u, \bar{S}_i \setminus \{j\})\, f_j(u)\, du\right]\right\}$$

- Weitzman reduces this to J static problems. Define the reservation utility z_{ij} by

$$c_{ij} = \int_{z_{ij}}^{\infty} (u - z_{ij})\, f_j(u)\, du.$$

  It is optimal to follow three rules:
  - **selection:** search options in decreasing order of z_{ij};
  - **stopping:** stop once the best utility found is at least the largest z_{ij} among unsearched options;
  - **choice:** buy the searched option with the highest utility.
- If $F_j = F$ and $c_{ij} = c_i$, the reservation utility is the same for every option, so the selection rule is silent: search is random.

Kim, Albuquerque & Bronnenberg (2010)

- Amazon "view rank" data: the commonality index of products j and k is $CI_{jk} = \frac{n_{jk}}{\sqrt{n_j}\sqrt{n_k}}$, where n_{jk} is joint viewership.
- Map it into a sequential search model. Order products by reservation utility; the product in position k is searched iff none of the $k-1$ products searched before it beat z_{ik}:

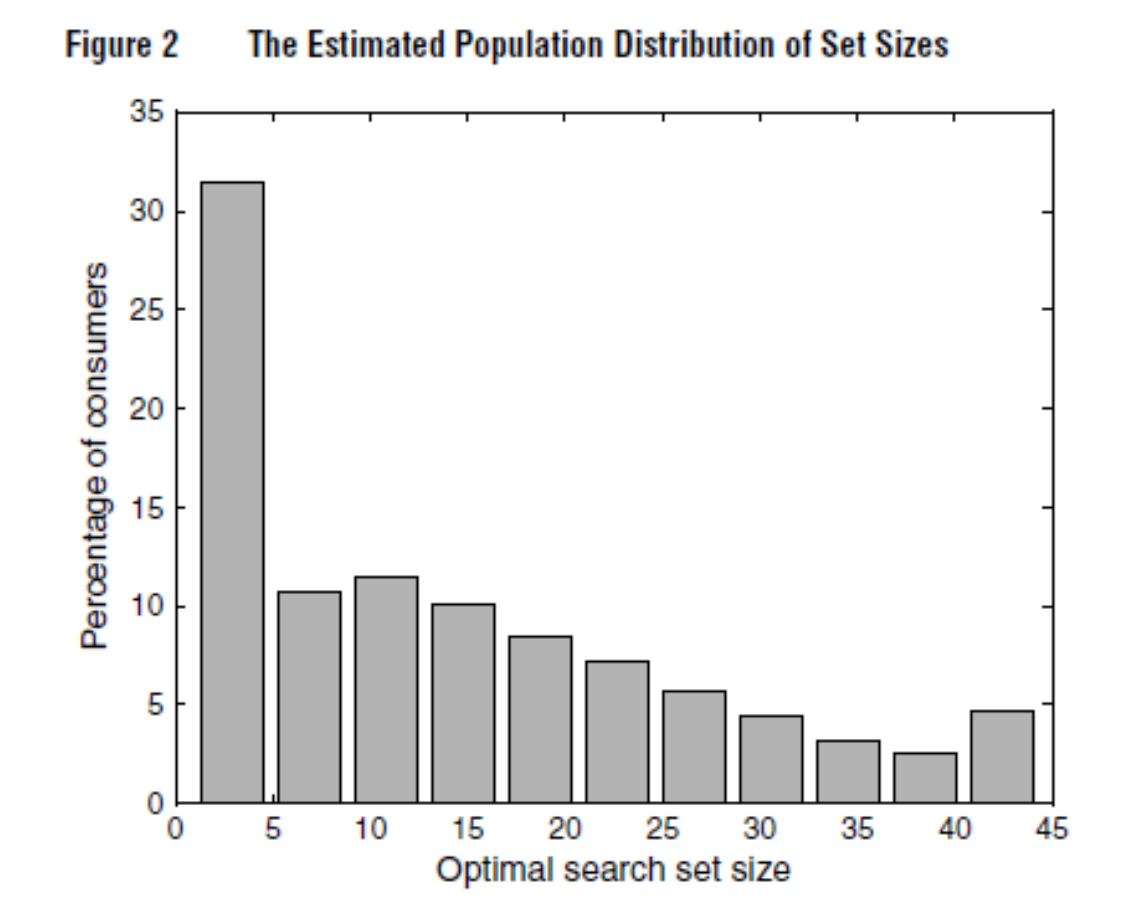
$$\pi_{ik} = \Pr\left(\max_{l=1,\ldots,k-1}\{\delta_l + \epsilon_{il}\} < z_{ik}\right) = \prod_{l=1}^{k-1} F_l(z_{ik} - \delta_l),$$

  with z_{ij} defined by $\int_{z_{ij}}^{\infty}(u - z_{ij})\, dF_j(u) = c_{ij}$.
- Summing over individuals gives $\hat{n}_j = \sum_i \pi_{ij}$.
- A search set that contains a lower-ranked product always contains the higher-ranked one, so $\hat{n}_{jk} = \sum_i \min(\pi_{ij}, \pi_{ik})$, and

$$\widehat{CI}_{jk}(\Theta, X) = \frac{\sum_i \min(\pi_{ij}, \pi_{ik})}{\sqrt{\sum_i \pi_{ij}}\sqrt{\sum_i \pi_{ik}}}.$$

- Amazon reports pairwise rankings of CI_{jk} vs. CI_{jl}; the structural parameters are chosen to rationalize these observed inequalities in a probit.

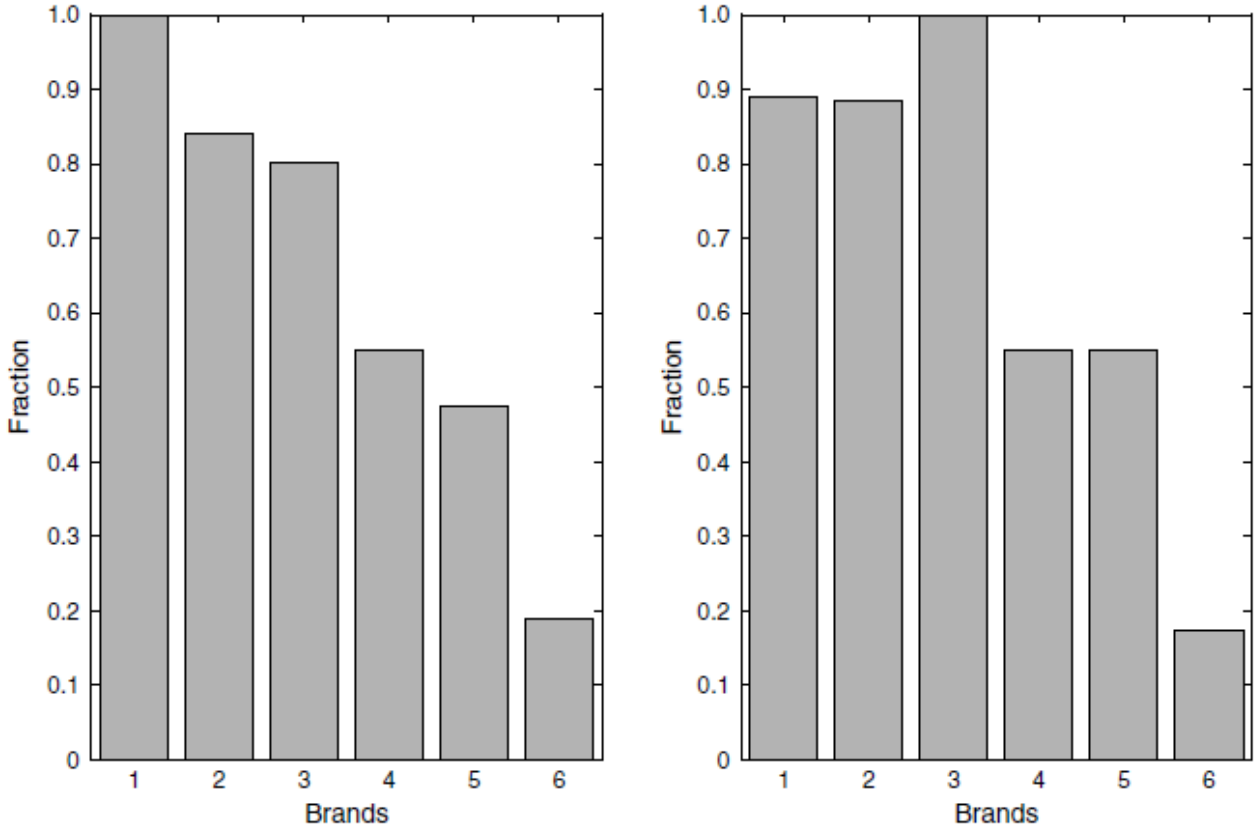
Kim et al. (2010): Implied Size of Search Sets



- Viewership (search-set) data alone identify both preferences and search costs, under the maintained assumption of optimal search.

Kim et al. (2010): Joint Search Shares

Figure 4 **Joint Search Share Conditional on Search on Sony (Left) and Canon (Right)**



Note. The brands are 1 (Sony), 2 (Panasonic), 3 (Canon), 4 (JVC), 5 (Samsung), and 6 (Sanyo).

Hortaçsu & Syverson (2004)

- Retail S&P 500 index funds: enormous fee dispersion (the 75th-to-25th percentile ratio is 3.1).
- The funds are financially homogeneous, so price differences must come from non-portfolio factors.
- Two candidate forces: investors' taste for non-portfolio attributes (vertical differentiation) and informational (search) frictions.
- Use equilibrium conditions to explain the data quantitatively.
- Policy: what if the market were restricted to a single fund? Less search vs. monopoly.

- Relative to Weitzman: options differ, but investors *cannot* direct their search; funds are sampled at random with probabilities ρ_j.

Hortaçsu & Syverson (2004): Industry

- Dispersion in returns is tiny: interquartile range 0.32 percentage points.
- A growing industry: the number of funds rises from 24 to 85 between 1995 and 2000.
- The high-price segment and return dispersion both increase over the years.
- Model: a continuum of investors.

Hortaçsu & Syverson (2004): Model

- $u_j = W_j\beta - p_j + \xi_j$, with the price coefficient normalized to -1.
- Investors' search costs are heterogeneous, $c \sim G(c)$; optimal stopping at reservation utility z^* such that

$$c = \int_{z^*}^{\bar{u}} (u - z^*)\, dH(u),$$

where investors know $H(u) = \frac{1}{N}\sum_{j=1}^{N} \mathbb{1}(u_j \leq u)$, with funds ranked $u_1 < u_2 < \cdots < u_N$.

- Let ρ_k be the probability that fund k is sampled. The *lowest* search cost of any investor who stops at fund j is

$$c_j = \sum_{k=j}^{N} (u_k - u_j)\rho_k, \qquad c_N = 0 < c_{N-1} < \cdots < c_1.$$

- Higher-quality funds are associated with lower search-cost cut-offs.

Hortaçsu & Syverson (2004): Shares and Pricing

- N market share equations: the lowest-utility fund $q_1 = (1 - G(c_1))\rho_1$; the second lowest

$$q_2 = \rho_2 \underbrace{(1 - G(c_1))}_{\text{stop at 1}} + \underbrace{\frac{\rho_2}{1 - \rho_1}}_{\text{sample 2| skip 1}} \underbrace{(G(c_1) - G(c_2))}_{\text{stop at 2}};$$

and in general, for $j = 3, \ldots, N$,

$$q_j = \rho_j \left[1 + \frac{\rho_1 G(c_1)}{1 - \rho_1} + \sum_{k=2}^{j-1} \frac{\rho_k G(c_k)}{(1 - \cdots - \rho_{k-1})(1 - \cdots - \rho_k)} - \frac{G(c_j)}{1 - \cdots - \rho_{j-1}} \right]$$

- N profit-maximization conditions:

$$q_j(\mathrm{p}, \mathrm{W}) + (p_j - mc)\, \partial q_j(\mathrm{p}, \mathrm{W}) / \partial p_j = 0$$

- **Both** prices and quantities are observed.

Hortaçsu & Syverson (2004): Identification

Given ρ_j for all j:

- The market share equations deliver the CDF at the cut-offs, $G(c_j)$, with $G(c_N) = 0$.
- The cut-offs themselves still need to be identified. With no vertical differentiation, $c_j = \sum_{k=j}^{N} \rho_k (p_j - p_k)$ comes directly from prices; with differentiation it is harder.
- The first-order conditions identify the density at the cut-offs, $g(c_j)$:

$$\frac{\partial q_j}{\partial p_j} = -\frac{\rho_1 \rho_j^2 \, g(c_1)}{1-\rho_1} - \frac{\rho_2 \rho_j^2 \, g(c_2)}{(1-\rho_1)(1-\rho_1-\rho_2)} - \cdots \equiv -\frac{q_j}{p_j - mc}$$

- Then apply the trapezoid rule:

$$G(c_{j-1}) - G(c_j) = \tfrac{1}{2}(c_{j-1} - c_j)\big(g(c_{j-1}) + g(c_j)\big);$$

  given $c_N = 0$, all cut-offs are identified.
- Finally, the u_j follow from $c_j = \sum_{k=j}^{N} (u_k - u_j)\rho_k$.

Hortaçsu & Syverson (2004): Estimation

Follows the identification argument closely, with more parametric structure:

- $\rho_j = \frac{Z_j^\alpha}{\sum_{k=1}^N Z_k^\alpha}$;
- $G(c)$ lognormal; estimate $c_1, \ldots, c_N$ by NLLS given $G(c_j), g(c_j)$;
- marginal cost is a parameter common to all funds;
- the search cost distribution is estimated separately for each year;
- utility from non-portfolio characteristics W_j is estimated with an IV for price.

Hortaçsu & Syverson (2004): Search Cost Estimates

TABLE III
SEARCH MODEL WITH UNEQUAL SAMPLING PROBABILITIES

Parameters	(A)	(B)	(C)	(D)	(E)
Log (mean search cost)	−6.17	−6.68	−6.58	−6.78	−6.33
	(0.06)	(0.06)	(0.13)	(0.26)	(0.11)
Variance of logged search costs	1.88	2.07	1.79	1.89	1.95
	(0.03)	(0.06)	(0.06)	(0.08)	(0.04)
Mean marginal cost, basis points	4	12	11	16	4
	(1)	(1)	(2)	(2)	(1)
α	2.62	2.43	2.58	2.44	3.06
	(0.04)	(0.03)	(0.03)	(0.09)	(0.04)
γ	—	—	—	—	0.11
					(0.03)
Time trend of mean search cost	−0.38	−0.50	−0.29	−0.51	−0.30
	(0.05)	(0.04)	(0.08)	(0.08)	(0.06)
Time trend of search cost variance	0.20	0.18	0.17	0.18	0.16
	(0.05)	(0.04)	(0.02)	(0.02)	(0.02)
R^2, prices	0.92	0.97	0.82	0.94	0.99
R^2, quantities	0.98	0.83	0.98	0.84	0.98
Median search cost (1996), b.p.	21	12	14	11	18
IQR of search cost range (1996), b.p.	5.9 to 75	3 to 50	4 to 46	3 to 41	4.7 to 67
Median search cost (2000), b.p.	5	2	4	1	5
IQR of search cost range (2000), b.p.	0.7 to 28	0.3 to 11	0.8 to 23	0.2 to 9	0.9 to 30

- Search costs are modest: about 5 basis points (5 per 10,000 of assets).
- Considerable heterogeneity, trending down over time.
- Substantial asymmetry in funds' sampling probabilities.

Hortaçsu & Syverson (2004): Non-portfolio Characteristics

TABLE IV
UTILITY FUNCTION ESTIMATES AND FUND ATTRIBUTE SUMMARY STATISTICS

Attribute	Utility weight, basis points (s.e.)	Mean	Standard deviation
Constant	93.53	N/A	N/A
	(69.94)		
Any load dummy	−12.11	0.547	0.499
	(33.74)		
Rear/deferred load dummy	59.57	0.272	0.446
	(37.13)		
Exchange-traded fund	199.5*	0.023	0.149
	(58.88)		
Number other share classes	2.726	1.621	1.337
	(9.722)		
log[no. funds in same mgmt. company]	30.97*	4.259	1.215
	(12.67)		
log[fund age]	99.39	1.393	0.728
	(54.81)		
Manager tenure (years)	3.578	2.922	2.776
	(11.03)		
Income + capital gains yield (percent)	−6.552*	3.248	3.363
	(3.009)		
Avg. monthly percent diff. between fund and S&P 500 returns	136.4*	−0.026	0.106
	(56.69)		
Standard deviation of monthly returns (percent)	48.22*	4.455	1.293
	(8.101)		
N	309		
R^2	0.354		
Mean of dependent variable	582.9		

Moraga-González, Sándor & Wildenbeest (2023): Motivation

- New cars in the Netherlands: to learn a car's match value you must visit a dealer (inspect, test drive), and dealers are mostly *single-brand*.
- Survey evidence: new-car buyers visit about 2 dealers on average; 47% of active searchers visit one brand only.
- Distance to the nearest dealer differs a lot across brands and households (e.g. 89% of households live within 7 km of an Opel dealer, 5% of a Rover dealer).

- In a full-information BLP model, a brand whose dealers are far away just looks *unattractive*: low shares are read as low ξ_j or high price sensitivity.

- Goal: put **optimal sequential (Weitzman) search** into a BLP-style demand system that
  - can be estimated from *aggregate* shares plus survey moments,
  - keeps a closed-form (logit-like) purchase probability,
  - delivers elasticities and markups for a standard Bertrand–Nash supply side.

MSW (2023): Model

- Utility of consumer i for car j

$$u_{ij} = \underbrace{\bar{\delta}_j + \alpha_i p_j + x_j' \beta_i}_{\delta_{ij}} + \varepsilon_{ij}, \qquad \bar{\delta}_j = \alpha p_j + x_j' \beta + \xi_j, \quad \varepsilon_{ij} \text{ i.i.d. EV1}$$

  with $\alpha_i = \tilde{\alpha}/y_i$ and random coefficients $\beta_i = \Pi D_i + \Sigma v_i$. Outside option $u_{i0} = \varepsilon_{i0}$.

- **Information.** Before searching, i knows (x_j, ξ_j, p_j), dealer locations, and ε_{i0}. Visiting firm f (selling the set G_f) at cost c_{if} reveals ε_{ij} for all $j \in G_f$.

- Search is over *firms*. The best car at f, $U_{if} = \max_{j \in G_f} u_{ij}$, is again EV1 with inclusive value

$$\delta_{if} = \log \sum_{h \in G_f} \exp(\delta_{ih}).$$

- Weitzman reservation value: with $H_{if}(r) = \int_r^\infty (z - r)\, dF_{if}(z)$, r_{if} solves $H_{if}(r_{if}) = c_{if}$, and

$$r_{if} = \delta_{if} + H_0^{-1}(c_{if}), \qquad H_0(r) = \int_r^\infty (z - r)\, dF_{\text{EV1}}(z).$$

  Better cars raise r_{if} one-for-one; higher search costs lower it.

MSW (2023): Search as Discrete Choice

- **Eventual purchase theorem** (Armstrong 2017; Choi, Dai & Kim 2018). Define the effective value

  $$w_{if} = \min\{r_{if},\ U_{if}\}.$$

  Following Weitzman's selection, stopping and choice rules, consumer i ends up buying from the firm with the *highest* w_{if} (with $w_{i0} = u_{i0}$).

- Intuition: f is searched before g iff $r_{if} > r_{ig}$; search stops at f iff U_{if} beats every remaining reservation value. Either way, only $\min\{r_{if}, U_{if}\}$ matters for whether f wins.

- So the dynamic search problem collapses to a *static* discrete choice over w:

$$s_{ij} = \underbrace{\frac{\exp(\delta_{ij})}{\exp(\delta_{if})}}_{P_{j|f}} \times \underbrace{\Pr\left(w_{if} \geq \max_{g \in \{0\} \cup \mathcal{F}} w_{ig}\right)}_{P_{if}}, \qquad P_{if} = \int \prod_{g \neq f} F_{w_{ig}}(z)\, dF_{w_{if}}(z).$$

- Since r_{if} falls with c_{if}, $F_{r_{if}}(z) = 1 - F_{if}^{c}\big(H_{if}(z)\big)$, and

  $$F_{w_{if}}(z) = 1 - \big(1 - F_{r_{if}}(z)\big)\big(1 - F_{if}(z)\big).$$

MSW (2023): Gumbel-Preserving Search Costs

- Choose the search cost distribution so that w_{if} is itself EV1:

$$F_{if}^{c}(c) = \frac{1 - \exp\left(- e^{-H_{\mathbf{0}}^{-\mathbf{1}}(c) - \mu_{if}} \right)}{1 - \exp\left(- e^{-H_{\mathbf{0}}^{-\mathbf{1}}(c)} \right)} \quad \Longrightarrow \quad w_{if} \sim \text{EV1 with location } \delta_{if} - \mu_{if}.$$

- Then the purchase probability is a logit with a search-cost penalty

$$s_{ij} = \frac{\exp(\delta_{ij} - \mu_{if})}{1 + \sum_{g \in \mathcal{F}} \sum_{k \in G_g} \exp(\delta_{ik} - \mu_{ig})}, \qquad s_j = \int s_{ij} \, dF(\tau_i).$$

- Search cost shifters

$$\mu_{if} = \log\left[1 + \exp(t_{if}' \lambda)\right] > 0,$$

  with t_{if}: a constant (plus a normal random constant), distance to the nearest dealer of f, log income, demographics, and distance $\times$ income.

- $F_{if}^{c}(0) = e^{-\mu_{if}}$: an atom of consumers who already know their match values at f.
- A random-coefficients version is a BLP model in which every firm carries a consumer-specific utility penalty μ_{if}; the contraction for $\bar{\delta}$ goes through unchanged.

MSW (2023): Identification

- The closed form makes the problem transparent: δ_{ij} and μ_{if} enter *additively*.
- **Aggregate shares alone.** The *level* of search costs is not identified: it is absorbed into the brand-level mean utilities. Only variation in search costs from shifters excluded from utility is identified.
- **Distance is the natural shifter**, but it can also enter utility (service, maintenance). Purchase data identify only its *net* effect, $\beta^{\delta}_{\mathsf{dist}} - \lambda_{\mathsf{dist}}$.
- **Search data break the tie.** Utility and search costs move reservation values and search behaviour differently: a more attractive car raises *both* the chance it is searched and the chance it is bought once searched; a higher search cost lowers search without changing the post-search purchase decision. Moments on the probability of visiting dealers (searching 0, 1, ≥ 2 times), including by non-buyers, separate the two.
- Survey support: in the raw data, distance lowers the probability of visiting a brand's dealer.

MSW (2023): Data and Estimation

- **Aggregate data**, Netherlands 2003–2008: 1,382 model-year observations (213–241 models per year), 39 brands owned by 16 companies; prices and characteristics (Autoweek), sales (BOVAG).
- **Dealer locations**: distance from each neighbourhood centroid to the nearest dealer of each brand.
- **Survey** (TNS NIPO, 2,530 respondents): brands of dealers visited, test drives, demographics, zip code.
- Estimation follows micro-BLP (Berry, Levinsohn & Pakes 2004; Petrin 2002):
  - contraction mapping for $\bar{\delta}_j$ from aggregate shares;
  - *purchase* micro moments by demographic group (price $\times$ income, size $\times$ kids, ...);
  - *search* micro moments: the model's probabilities of not searching and of searching once, matched to the survey, by search-cost covariate;
  - price instruments: unit labour costs, steel price $\times$ weight, model-level production; average dealer distance as an additional excluded shifter.
- **Supply**: multi-product Bertrand–Nash, $p - mc = \Delta(p)^{-1} s(p)$, with Δ built from the search-model derivatives.

MSW (2023): Search Changes Elasticities and Markups

Conditional logit specifications (working-paper version, CEPR DP 16574):

	Price coef.	Own-price elast.	Markup
Full-information logit	−0.168	−4.91	3.71
Search logit (Gumbel-preserving)	−0.130	−3.81	4.81
Search logit (normal costs)	−0.130	−3.80	4.84
Full-information nested logit	−0.132	−5.41	4.71

- Distance significantly raises search costs (coefficient 0.18 in the search-cost index, s.e. 0.04), and the result is robust to the search-cost distribution (Gumbel-preserving vs. normal).
- Why demand looks *less* elastic once search is modelled: expensive models tend to be sold by dealers farther from consumers. Without search costs, their low shares are attributed to price sensitivity.
- Cross-effects: when a brand's dealers are far away, consumers substitute mostly *within* brand and toward similar brands. Proximity, not only characteristics, shapes substitution.

MSW (2023): Exclusive Dealing

- Policy question: car dealers are mostly *exclusive* (single-brand). What if they were multi-brand?
- Mechanism: one visit then reveals the match values of several brands, which lowers effective search costs and raises the number of products consumers compare.
- Counterfactual findings
  - A single brand that unilaterally sells through multi-brand dealers gains: prices fall, but the brand is searched more and its profits rise.
  - If the whole industry moves to multi-brand dealers, competition intensifies, prices fall and *industry* profits fall substantially: a prisoner's dilemma.
  - Manufacturers have little incentive to combine even brands within their own group.
  - So exclusive dealing benefits firms at consumers' expense: consumers face higher search costs and higher prices.
- These effects are much larger than a full-information model would predict, because there market structure changes only through characteristics, not through consideration.

Summary (I): Economics

- **Consideration sets** (Goeree 2008): limited awareness, shifted by advertising; identified from functional form plus micro data on media exposure.
- **Search frictions are a source of market power** (Diamond 1971) and, with heterogeneous information, a source of equilibrium price dispersion (Burdett–Judd 1983).
- **Protocol matters.** Simultaneous search needs a ranking result (Chade–Smith 2006); sequential search has Weitzman's index rule, and the eventual purchase theorem turns it into a static discrete choice problem.
- **For applied IO**: ignoring search tends to mismeasure elasticities and markups, and misses counterfactuals that work through consideration (exclusive dealing, platform design, advertising).

Summary (II): What the Data Must Contain

- *Hong & Shum (2006)*. Prices only $\Rightarrow$ search cost distribution from firm indifference.
- *Hortaçsu & Syverson (2004)*. Prices and quantities $\Rightarrow$ search costs and quality jointly.
- *Honka et al. (2017)*. Awareness and consideration sets $\Rightarrow$ advertising's awareness vs. utility effects.
- *Kim et al. (2010)*. View / co-search data $\Rightarrow$ preferences and search costs from search alone.
- *Moraga-González et al. (2023)*. Aggregate shares + search moments + distance $\Rightarrow$ search costs inside BLP.
- Common thread: the *level* of search costs is never free; it is pinned down by equilibrium restrictions (price data) or by observing search itself.

References (I)

- Goeree, M. S. (2008). "Limited Information and Advertising in the U.S. Personal Computer Industry." *Econometrica* 76(5), 1017–1074.
- Hong, H., and M. Shum (2006). "Using Price Distributions to Estimate Search Costs." *RAND Journal of Economics* 37(2), 257–275.
- Honka, E., A. Hortaçsu, and M. A. Vitorino (2017). "Advertising, Consumer Awareness, and Choice: Evidence from the U.S. Banking Industry." *RAND Journal of Economics* 48(3), 611–646.
- Hortaçsu, A., and C. Syverson (2004). "Product Differentiation, Search Costs, and Competition in the Mutual Fund Industry: A Case Study of S&P 500 Index Funds." *Quarterly Journal of Economics* 119(2), 403–456.
- Kim, J. B., P. Albuquerque, and B. J. Bronnenberg (2010). "Online Demand under Limited Consumer Search." *Marketing Science* 29(6), 1001–1023.
- Moraga-González, J. L., Z. Sándor, and M. R. Wildenbeest (2023). "Consumer Search and Prices in the Automobile Market." *Review of Economic Studies* 90(3), 1394–1440.

References (II)

- Armstrong, M. (2017). "Ordered Consumer Search." *Journal of the European Economic Association* 15(5), 989–1024.
- Burdett, K., and K. L. Judd (1983). "Equilibrium Price Dispersion." *Econometrica* 51(4), 955–969.
- Chade, H., and L. Smith (2006). "Simultaneous Search." *Econometrica* 74(5), 1293–1307.
- Choi, M., A. Y. Dai, and K. Kim (2018). "Consumer Search and Price Competition." *Econometrica* 86(4), 1257–1281.
- Diamond, P. A. (1971). "A Model of Price Adjustment." *Journal of Economic Theory* 3(2), 156–168.
- Honka, E., A. Hortaçsu, and M. Wildenbeest (2019). "Empirical Search and Consideration Sets." In *Handbook of the Economics of Marketing*, Vol. 1, 193–257.
- Stigler, G. J. (1961). "The Economics of Information." *Journal of Political Economy* 69(3), 213–225.
- Weitzman, M. L. (1979). "Optimal Search for the Best Alternative." *Econometrica* 47(3), 641–654.